

|   |                                                                                                            |    |     |     |
|---|------------------------------------------------------------------------------------------------------------|----|-----|-----|
| 5 | (a) wave incident on/passes by or through an aperture/edge<br>wave spreads (into geometrical shadow)       | B1 | B1  | [2] |
|   | (b) $n\lambda = d \sin \theta$                                                                             | C1 |     |     |
|   | substitution of $\theta = 90^\circ$ or $\sin \theta = 1$                                                   | C1 |     |     |
|   | $4 \times 500 \times 10^{-9} = d \times \sin 90^\circ$                                                     |    |     |     |
|   | line spacing = $2.0 \times 10^{-6} \text{ m}$                                                              | A1 | [3] |     |
|   | (c) wavelength of red light is longer (than 500 nm)                                                        | M1 |     |     |
|   | (each order/fourth order is now at a greater angle so) the fifth-order maximum cannot be formed/not formed | A1 | [2] |     |